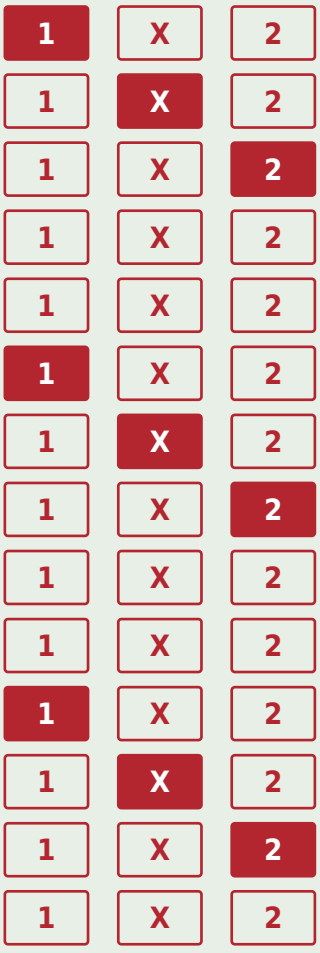
Family Quiniela

How it all works

Complete guide to the automated system: what it does, how it is used, how it is built and how it is maintained.



Luis Angel · Rosana · Joseba · Lara · Borja

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1. In a nutshell

Every week my family (Luis Angel, Rosana, Joseba, Lara and Borja) plays the **Quiniela**, Spain's official football pools: each of us predicts the outcome of 14 matches (1 = home win, X = draw, 2 = away win) plus a 15th bonus match. One ticket is stamped at a lottery shop with everyone's column, and when the matchday ends **whoever gets the fewest hits pays the next week's ticket** (€3.75).

It used to be entirely manual. Now a **bot** (a program that runs on its own) takes care of it all:

- It detects when the lottery publishes the week's LaLiga Quiniela and announces it in the Telegram group.
- Each player fills in their column on a **web app** from their phone, using a PIN.
- It reminds whoever has forgotten, 48 hours before the first kickoff.
- Once the five columns are in, it posts an image of the ticket, ready to be stamped.
- While the matches are played, it posts how everyone is doing at the end of each day.
- At the end it announces who pays next and, if any column has won a prize, how much.

Only what genuinely needs a person is still manual: **filling in the column**, **stamping the ticket** at the lottery shop and **paying**. **Everything runs on free services** and nobody has to watch over it.

Goals

- **Remove the chores:** nobody copies columns, checks results or does the maths.
- **Avoid mistakes:** hits and payments are computed by the program from the official results.
- **Nobody forgets:** automatic announcement and a reminder to whoever is missing.
- **More fun:** daily standings while the matchday is being played.
- **A permanent record:** every matchday, column, payment and prize is stored.

Benefits

Before	Now
Someone had to share the link every week.	The bot announces the matchday as soon as the lottery publishes it.
Everyone forwarded their prediction to the group by hand.	It is entered on the web app and saved automatically.
Someone had to chase whoever was missing.	Automatic reminder 48 h before, mentioning whoever is missing.
Copying the five columns to stamp the ticket.	An image with the five columns, ready for the lottery shop.
Checking results, counting hits and deciding who pays.	Hits in green and misses in red, day by day, and the bot says who pays (splitting ties).
Checking whether any column won a prize.	A message with each column's prize from the official breakdown.

2. A week with the system

This is how a normal weekend matchday goes. None of these steps needs anyone to press anything, except filling in their column. The players' messages are in Spanish; translations are shown here.

When	What happens	Who
When the lottery publishes the matchday (usually midweek)	Group message *«Ya está la quiniela de la jornada 16»* (“Matchday 16 is out”) with an image of the 15 fixtures, the deadline, who fills in the Pleno al 15, who pays for this ticket and a **Hacer mi quiniela** (“Fill in my column”) button.	Bot
Until 24 h before the first kickoff	Each player opens the web app, picks their name, enters their PIN, marks the 14 matches and submits. It can be changed as many times as needed until the deadline.	Each player
48 h before the first kickoff	If someone is missing, a reminder in the group mentioning them.	Bot
When all five are in (or betting closes)	Image with the five columns and who pays for the ticket: *«Joseba debe abonarla con 3,75 €»*.	Bot
Before the first kickoff	Go to the lottery shop, stamp the ticket with those columns and pay for it.	Whoever pays
After the last match of each day	Progress image: hits in green, misses in red and who is leading.	Bot
After the last match of the matchday	Final image, everyone's hits and who pays the next ticket.	Bot
When the lottery publishes the prize breakdown	If any column won: *«Borja ha tenido 11 aciertos por lo que corresponde 2,22 € según el sorteo»* (“Borja got 11 hits, which pays €2.22”).	Bot

A normal week brings 5 or 6 messages. The progress report is **one per match day**, not one per match, so the group is never flooded.

What it looks like

Jornada 14

Así va tras el sábado · 7 de 14 jugados

	Partido	Luis	Rosana	Joseba	Lara	Borja	Res.
1	Athletic Club - Valencia	1	1	1	1	1	1
2	Real Madrid - Sevilla	1	X	X	1	1	X
3	Betis - Barcelona	2	2	1	2	2	2
4	Osasuna - Getafe	2	1	X	X	1	1
5	Villarreal - Alavés	1	1	1	1	1	1
6	Celta - Mallorca	X	1	X	X	X	X
7	Girona - Rayo Vallecano	1	X	1	1	1	X
8	At. Madrid - Real Sociedad	1	1	X	X	X	
9	Espanyol - Levante	1	1	1	1	1	
10	Elche - Oviedo	X	2	X	X	X	
11	Racing - Deportivo	1	1	1	1	1	
12	Eibar - Zaragoza	1	X	X	1	1	
13	Sporting - Málaga	X	2	1	2	2	
14	Almería - Granada	1	1	X	1	1	
15	Real Sociedad - Athletic					2-1	
	Aciertos	4	6	4	4	5	

Jornada 14 · final

Resultado oficial

	Partido	Luis	Rosana	Joseba	Lara	Borja	Res.
1	Athletic Club - Valencia	1	1	1	1	1	1
2	Real Madrid - Sevilla	1	X	X	1	1	X
3	Betis - Barcelona	2	2	1	2	2	2
4	Osasuna - Getafe	2	1	X	X	1	1
5	Villarreal - Alavés	1	1	1	1	1	1
6	Celta - Mallorca	X	1	X	X	X	X
7	Girona - Rayo Vallecano	1	X	1	1	1	X
8	At. Madrid - Real Sociedad	1	1	X	X	X	1
9	Espanyol - Levante	1	1	1	1	1	2
10	Elche - Oviedo	X	2	X	X	X	X
11	Racing - Deportivo	1	1	1	1	1	1
12	Eibar - Zaragoza	1	X	X	1	1	X
13	Sporting - Málaga	X	2	1	2	2	2
14	Almería - Granada	1	1	X	1	1	1
15	Real Sociedad - Athletic					2-1	2-1
	Aciertos	8	11	7	8	9	

Left: Saturday progress report (7 of 14 matches played). Right: final result. Green = hit, red = miss. Images from a test matchday.

3. Game rules

Who pays

- The player with **the fewest hits** pays the next ticket (€3.75).
- Ties at the bottom **split the price**: with two, €1.87 and €1.88 (leftover cents go to one of them); with three, €1.25 each.
- A player who does not submit a column before the deadline counts as **0 hits**.

Pleno al 15

- The 15th match, where each team's goals are predicted: 0, 1, 2 or M (3 or more).
- It is filled in by **the best player of the previous matchday**. Ties among the best are drawn at random; the very first matchday was drawn among all five.
- It does not add hits for deciding who pays. It only matters if someone gets all 14 right.
- The official ticket has **a single Pleno for all columns**, so if any column gets 14 hits and the Pleno is also right, that special prize belongs to that column.

Deadlines

- Columns can be submitted and changed until **24 hours before the first kickoff**. After that the web app locks them.
- The reminder goes out **48 hours before** the first kickoff, only to whoever has not submitted.
- If the lottery publishes a matchday with less than 24 h to go, the bot skips it: it could no longer be played.

Postponements and rescheduling

- Until the matchday starts, the bot re-reads the fixtures on every run. If a name or kickoff time changes, the web app and the images update themselves.
- If the **first kickoff** moves while betting is still open, the deadline moves with it and the bot tells the group (*«Cambio de horario en la jornada 15…»*).
- If a match is postponed or suspended, that day's report still goes out 8 hours after the day's last kickoff, naming the missing match. The final result waits for the official sign assigned by the lottery.

When a new matchday is announced

- Players bet on one matchday at a time, and the Pleno and who pays come from the previous one. So if the lottery publishes a matchday early, the bot waits: it is never announced while another is still open for betting.
- And if the previous one will finish before the new one closes, it waits until it finishes (unless less than 48 hours would be left to bet). Example: next weekend's matchday, published on a Wednesday, is announced on Monday night when this weekend's ends, already with the right Pleno player and payer.

Two matchdays at once

Example: the weekend matchday ends on Monday at 23:00 and the next one (midweek) starts on Tuesday at 18:00. Then:

- Betting on the midweek one closes on **Monday at 18:00** (24 h before its first kickoff), before the other ends.
- Its **Pleno al 15** is assigned using the last finished matchday, as the weekend one has no winner yet.
- **Who pays** is known on Monday night, when the weekend matchday ends; that person stamps the ticket on Tuesday.
- The two Quinielas are fully separate and every message carries its matchday number.

Which Quinielas are played

- Only **LaLiga** ones: those with at least 5 matches between two top-flight (Primera División) teams. Champions League, Copa del Rey and other competitions are skipped.
- Midweek LaLiga rounds count too.
- **Women's** teams are shown with **(F)** (e.g. «Real Madrid (F) - Tenerife (F)») and never count for this filter.
- The list of top-flight teams is downloaded automatically every season, so promotions and relegations need no changes.

Prizes

When the lottery publishes the official prize breakdown, the bot checks the five columns and posts one line per winning column, plus the total if there are several:

```
Borja ha tenido 11 aciertos por lo que corresponde 2,22 € según el sorteo.
```

If no column won, nothing is sent. The categories are: Pleno al 15, 14, 13, 12, 11 and (some weeks) 10 hits.

4. For the players: how to bet

Nothing to install. Everything happens on a web page from the phone (in Spanish).

Step	What to do
1	Always open it from the **Hacer mi quiniela** button in the bot's message: it carries the family key. The phone remembers it afterwards.
2	Tap your name and type your 4-digit **PIN**, sent privately by Borja.
3	Mark one sign (1, X or 2) for each of the 14 matches. If it is your turn for the Pleno al 15, mark the goals too.
4	Tap **Registrar mi quiniela** (“Submit”). A confirmation appears and a ✓ next to your name.
5	Made a mistake? Open it again, change anything and submit again, until 24 h before the first kickoff.

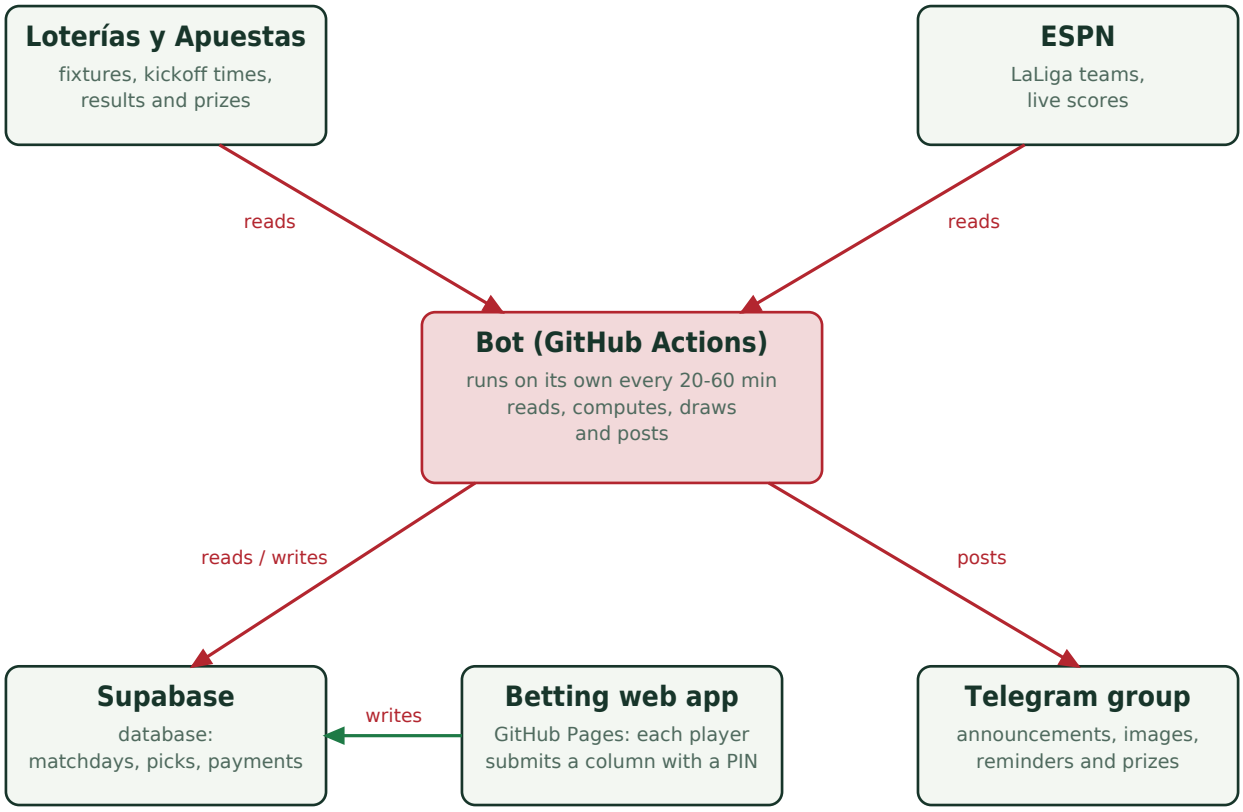


The web app: picking a name and PIN (left) and filling in the column (right). Test screenshot.

Why a PIN? Everyone uses the same link. The PIN makes sure each player can only edit **their own** column, so nobody changes someone else's by accident. After 5 wrong attempts that name is locked for 15 minutes.

5. How it is built

The system is made of these components. None of them needs a computer switched on at home: everything runs on free cloud services.



How they fit together: the bot reads data from the lottery and ESPN, stores and queries everything in Supabase, and posts to Telegram. The web app only talks to Supabase, to save the columns.

Component	Service	Role	Cost
Code	GitHub (public repository)	Holds the whole program and its change history.	Free
Bot	GitHub Actions (Python 3.12)	Runs every 10 minutes, started by cron-job.org through the GitHub API (GitHub's own schedule is kept as a fallback).	Free and unlimited for public repos
Web app	GitHub Pages (HTML + JavaScript)	Page where each player fills in their column.	Free
Database	Supabase (PostgreSQL)	Matchdays, fixtures, picks, reports sent and payments.	Free plan: 500 MB; a few KB are used
Messaging	Telegram Bot API	Posts images and messages to the group.	Free
Fixture data	Loterías y Apuestas del Estado	Official fixtures, kickoff times, results and prize breakdown.	Free (public website)
Teams and live scores	ESPN	Current season's top-flight teams, and each final score minutes after the match.	Free, no sign-up
Backup	GitHub Actions	Every Tuesday stores an encrypted copy of the database in the repository.	Free

What the bot does on every run

Each run (a *tick*) does the same thing and **never repeats a message**: every message is flagged in the database once sent (announced, reminder sent, picks published, day X reported, finished, prizes published). The design is idempotent, so it does not matter how often it runs.

- **1. Find new matchdays.** Asks the lottery for upcoming draws. For each new one, checks it is a LaLiga Quiniela (using the ESPN team list) and that there is still time to bet. If so, assigns the Pleno and stores it.
- **2. Process every open matchday.** Refreshes names and kickoff times, announces it if needed, reminds whoever is missing, publishes the columns and, once started, loads results: the official ones from the lottery and, for matches it has not published yet, the final score from ESPN.
- **3. Daily reports and closing.** When all of a day's matches have a result, posts that day's report. When all 14 are in, closes the matchday, works out who pays and stores it.
- **4. Prizes.** For finished matchdays, waits for the official prize breakdown and reports any winning column.

Separately, every **Tuesday** another workflow takes an **encrypted backup** of the whole database (see section 7).

There can be **two matchdays at once** (e.g. a midweek LaLiga round open for betting while the weekend one is still being played). **They never mix:** each has its own announcement, deadline, columns, reports and ticket, and the new one's announcement says so explicitly. The web app always shows the one open for betting.

6. Technical details

Where the data comes from

The lottery (LAE) has no documented public API. The bot uses the same internal JSON endpoints as its website:

Endpoint (loteriasyapuestas.es/servicios/...)	Purpose
proximosv3?game_id=LAQU&num=3	Upcoming draws: matchday number, draw date, draw id and official close.
fechav3?game_id=LAQU&fecha_sorteo=YYYYMMDD	A draw's 15 fixtures, in ticket order, with date and time.
buscadorSorteos?game_id=LAQU&celebrados=true&...	Settled draws: official results and prize breakdown per category.

The lottery returns HTTP 403 to clients that do not look like a browser, so the bot connects through **curl_cffi**, which impersonates Chrome's TLS fingerprint. Teams arrive as «Athletic Club (m)» or «Real Madrid (f)»: the (m) is dropped and (f) becomes (F). All parsing is tolerant of format variations, and a **diagnose** command prints exactly how the data is being interpreted.

ESPN provides the top-flight team list. Since both sources spell names differently («At. Madrid» vs «Atlético Madrid», «Betis» vs «Real Betis»), names are compared by their meaningful words, one way only, so reserve teams («Sevilla At.», «Real Madrid Castilla») never match the first team. If ESPN is down, a fallback list is used.

ESPN also supplies **live scores**. The lottery only publishes results once the whole draw is settled (Monday night or later), so without them no daily report could be sent. ESPN's final scores (Primera and Segunda) are matched to the ticket by kickoff time and team names, and the lottery's official results replace them as soon as they appear.

Schedule. GitHub runs scheduled workflows late when it is busy (hours, in practice). So the free service **cron-job.org** starts the bot every 10 minutes through the GitHub API, using a token that can only run this repository's workflows; GitHub's own schedule remains as a fallback. Runs never overlap and nothing is sent twice.

The database

Code, database and technical documentation are in English. Everything the players see (messages, images and the web app) is in Spanish, isolated in `bot/messages.py` and the web page.

Table	Contents
players	The five players, their hashed PIN and their Telegram id (for mentions).
matchdays	Matchday number, season, lottery draw id, first kickoff, Pleno player and the flags of what has been sent.
matches	Each matchday's 15 fixtures with kickoff, goals and sign once played.
picks	Each player's column (14 signs) and the Pleno, with the last update time.
daily_reports	Days whose progress report has been sent, so it is never repeated.
payments	Who lost each matchday and how much they owe for the next ticket.

Security

- **Family key:** the repository and the website are public, so the website only works with a secret key that travels inside the link of the bot's «Hacer mi quiniela» button in the Telegram group. It is never in the repository. Anyone opening the site another way only sees «Quiniela privada», even if they know a PIN. The key lives in the URL fragment, which browsers never send to any server; the phone remembers it and it is removed from the address bar.
- **Hashed PINs:** stored with bcrypt (an irreversible hash), never in plain text. After 5 failures in a row the player is locked for 15 minutes.
- **The web app cannot touch tables:** it can only call three database functions (public state, your own column and submit your column), all of which require the family key, and the column ones the PIN too. Everything else, including changing a PIN, is denied (PostgreSQL's default PUBLIC execute grant is revoked). Tables are protected with Row Level Security.
- **Secrets out of the code:** the Supabase service key, the bot token and the group id are stored as GitHub Secrets, never visible in the code or the logs.
- **Encrypted backups:** the repository is public, so the weekly backup is encrypted (AES-256, PBKDF2) with a password that only exists in GitHub Secrets and in Borja's hands.
- **If the bot token leaks:** /revoke in @BotFather issues a new one and the old one stops working immediately.

Repository structure

Path	Contents
.github/workflows/quiniela.yml	“Bot” workflow: schedule and secrets.
.github/workflows/web.yml	“Deploy web” workflow: publishes the web app to GitHub Pages on every change.
.github/workflows/backup.yml	“Backup” workflow: weekly encrypted database backup.
bot/main.py	Orchestration: one idempotent run (tick) and the command-line interface.
bot/rules.py	Game rules: hits, who pays, Pleno, daily reports, league filter and prizes.
bot/lae.py	Client for the lottery's data (fixtures, results, prize breakdown).
bot/teams.py	LaLiga team list from ESPN and fuzzy name matching.
bot/messages.py	Every user-facing text (Spanish), in one place.
bot/ticket.py	Renders the ticket images (Pillow).
bot/db.py · bot/telegram.py	Supabase (PostgREST) and Telegram Bot API clients.
bot/backup.py	Database export and restore.
web/index.html · web/config.js	The betting web app and its Supabase connection.
supabase/schema.sql	Builds the whole database from scratch: tables, security, functions.
supabase/migrations/	One-off scripts already applied to the production database.
tests/test_simulation.py	Offline simulation of whole matchdays, including edge cases.
docs/	This guide, in English and Spanish.
backups/quiniela.json.enc	Latest encrypted backup; git history keeps the previous ones.

7. Maintenance and troubleshooting

The system is designed to need no maintenance. Still, these are worth knowing:

Things that can happen

Situation	What to do
GitHub emails that a run failed.	GitHub → Actions → the red run → bot → Run, and read the error. Most likely the lottery changed its website.
The bot stops posting in the group.	If the group became a “supergroup” (happens when someone is made admin), its id changes to one starting with -100. Update the TELEGRAM_CHAT_ID secret.
Someone new joins, or someone's Telegram id changes.	Get their id with getUpdates and store it with an update on the players table.
A link with the family key was shared outside the family.	In Supabase: `select set_family_key();`, store the value as the FAMILY_KEY secret and re-send the matchday announcement. The old key stops working.
A PIN needs changing.	Supabase → SQL Editor: `select set_pin(3::smallint, '4821');`
Supabase pauses the project for inactivity.	It should not happen, as the bot uses it daily. If it does, it is resumed from the dashboard with no data loss.
GitHub disables the schedule for inactivity (60 days without commits).	It does not happen: the weekly backup commits every Tuesday (and the bot has a 50-day safety net).
git push says the local copy is out of date.	Because of the bot's weekly backup commits. Run `git pull` before `git push`.
GitHub warns that an action is deprecated.	Workflows already use the Node.js 24 versions. When a new one appears, bump the version in the .yml.

Backups

Supabase's free plan keeps no backups, so every Tuesday the **Backup** workflow exports all tables, encrypts them with the **BACKUP_KEY** secret and commits them to `backups/`. Restoring needs that password:

```
openssl enc -d -aes-256-cbc -pbkdf2 -iter 200000 -in backups/quiniela.json.enc -out backup.json
python -m bot.backup restore backup.json
```

Important: keep the BACKUP_KEY password somewhere outside GitHub too (e.g. a password manager). GitHub never shows a secret again, and without it the backup cannot be opened.

Useful operations

Run the bot by hand: GitHub → Actions → Bot → **Run workflow** → **tick**. To see what is being read from the lottery and ESPN: **Run workflow** → **diagnose**.

Re-send a matchday's announcement (e.g. if the message was deleted):

```
update matchdays set announced = false where number = 16;
```

Known limitations

- **Depends on the lottery's website.** It is not an official API: if it changes, bot/lae.py must be adapted.
- **Live scores depend on ESPN** (unofficial too). If ESPN fails, daily reports wait for the lottery's official results; the final result and who pays are always based on the official ones.
- **The cron-job.org token expires** (at most a year). GitHub emails a reminder; until it is renewed the bot falls back to GitHub's slower schedule.
- **The bot does not stamp the ticket** or check that it was stamped: that is still up to whoever goes to the shop.
- **Pleno al 15 with 14 hits:** the bot reports the special prize from the official breakdown. Should it happen, confirm at the lottery shop whether it adds to the 14-hit prize.
- **Overlapping matchdays:** the new one's Pleno is assigned from the last finished matchday, since the previous one is still in play.

8. Glossary

Term	Meaning
Quiniela	Spain's official football pools, run by Loterías y Apuestas del Estado (LAE).
Matchday (jornada)	One Quiniela round: 14 matches plus the Pleno al 15.
1 / X / 2	Predicted outcome of a match: home win, draw, away win.
Pleno al 15	The 15th match, where each team's goals are predicted (0, 1, 2, M). Special prize with 14 hits plus the Pleno.
Prize breakdown (escrutinio)	The lottery's official count of winners and the prize for each category.
Bot	A program that does tasks on its own. Here, the one that reads the lottery data and posts to Telegram.
Repository (repo)	A GitHub folder with all the code and the history of every change.
GitHub Actions / workflow	GitHub's service that runs programs on its servers on a schedule; a workflow is one such job.
GitHub Pages	GitHub's free static website hosting.
Supabase	Cloud PostgreSQL database service with a free plan.
Idempotent	Running it again has no extra effect: nothing is duplicated.
Secret	Private value (key, token) stored in GitHub, never visible in the code.
Token	A program's password. The bot token allows posting to the group as the bot.
PIN	Each player's 4-digit code to edit only their own column.
Overlapping matchdays	When a new matchday is open for betting while the previous one is still being played.
Commit / push	Saving a change to the history (commit) and uploading it to GitHub (push).

Setup steps and further technical detail are in the repository's README.md.